

Interview Questions — GraphQL & Next.js

Total Questions: 42

1. What is GraphQL, and how is it different from REST APIs?

Hint: Think about how data is requested and returned in both approaches.

2. Explain the roles of Schema, Query, Mutation, and Resolver in GraphQL.

Hint: Each term is a building block in how GraphQL defines and handles data.

3. What are two main benefits of using GraphQL over traditional REST APIs?

4. How does GraphQL allow the frontend to request only the data it needs from the backend?

5. Using the public GraphQL Pokémon API, how would you write a query to fetch the name and image of a specific Pokémon?

Hint: Focus on the 'query' keyword and specifying fields inside curly braces.

6. What is the purpose of the Apollo Client in a React application?

Hint: Consider how Apollo interacts with your React components and the GraphQL server.

7. Describe the steps to set up Apollo Client in a new React project.

8. What does the `Provider` component do in a React app using Apollo Client?

Hint: Think about React Context and how data is made available to components.

9. How do you connect your React app to a GraphQL endpoint using Apollo Client?

10. How does the `useQuery` hook work in Apollo Client, and what does it return?

11. Write a simple React component using `useQuery` to fetch and display a list of user names from a GraphQL endpoint.

Hint: Remember to handle loading and error states using the destructured values from `useQuery`.

12. How would you handle loading and error states when fetching data with `useQuery` in a React component?

13. Explain how you would fetch and render a list of users with their names and emails in a React component using GraphQL.

14. What is the difference between a GraphQL query and a mutation?

Hint: Consider what each is used for in terms of data operations.

15. How do you use the useMutation hook in Apollo Client to add a new user?

Hint: Think about the mutation definition and how the hook is used in a component.

16. Describe what is meant by 'Optimistic UI updates' in the context of GraphQL mutations.

17. Suppose you add a new user with a mutation. How does Apollo Client ensure the UI updates automatically without a page reload?

Hint: Consider how Apollo manages its local cache.

18. What does a simple GraphQL type definition like 'type User { id: ID! name: String! }' mean?

Hint: Focus on the meaning of the type, field names, and exclamation marks.

19. How can you use Apollo Sandbox or GraphiQL Playground to test custom queries and explore a GraphQL schema?

20. Write a nested GraphQL query to fetch a user's name and all their blog post titles.

Hint: Think about how to request related data in a single query.

21. Explain the difference between Server-Side Rendering (SSR) and Client-Side Rendering (CSR) in the context of Next.js.

Hint: How is the HTML generated and sent to the browser in each case?

22. What are some advantages of using Next.js over Create React App?

Hint: Think about rendering, routing, and SEO.

23. Describe the default folder structure of a new Next.js app. Which folder is responsible for routing?

Hint: Focus on the /pages directory.

24. How does SSR in Next.js help improve SEO and initial load performance?

25. How does file-based routing work in Next.js?

Hint: Consider what happens when you add a new file to the `/pages` directory.

26. What is a dynamic route in Next.js, and how would you create one for a blog post detail page?

Hint: Look at file naming conventions like `[id].js`.

27. How do you navigate between pages in a Next.js app?

Hint: Which component do you use for client-side navigation?

28. Explain how automatic routing and static path generation work in Next.js.

29. What is the difference between `getStaticProps`, `getServerSideProps`, and `getStaticPaths` in Next.js?

Hint: Consider when each function is executed and what kind of data fetching it supports.

30. When would you use `getStaticProps` instead of `getServerSideProps` in a Next.js project?

31. Describe how you would fetch blog post data at build time using `getStaticProps` and display it on a Next.js page.

32. How can you combine GraphQL queries with Next.js data fetching methods like `getStaticProps`?

Hint: Think about where you would place the GraphQL query and how the data is passed to the page.

33. What are environment variables in Next.js, and how do you use them securely?

Hint: Consider the `.env.local` file and why you shouldn't commit it.

34. How would you add authentication to a Next.js app using Firebase or NextAuth.js?

35. Explain how to protect API routes in Next.js so that only authenticated users can access them.

36. Describe the steps to add Google login to a Next.js application using NextAuth.js.

37. How would you handle protected pages in Next.js, ensuring a user is redirected if not logged in?

Hint: Think about session validation and redirect logic.

38. In the mini-project 'AI-Powered Blog Platform', how would you integrate OpenAI's API to generate article summaries?

Hint: Consider how you'd call the API and display the result in your blog platform.

39. If you are building a blog platform with Next.js, GraphQL, and authentication, which parts of your stack would handle data fetching, user sessions, and UI rendering?

40. Describe how you would use both GraphQL mutations and Next.js dynamic routes to allow users to edit blog posts in your project.

41. How could you use an AI tool like ChatGPT or GitHub Copilot to help you write a GraphQL query or debug a Next.js data fetching issue, and how would you verify the output?

Hint: Think about how AI can speed up your workflow, but also about checking for correctness.

42. What are some ways AI tools like OpenAI or Copilot can support your work in building a React, Next.js, or GraphQL project, and what tasks would still require manual review or testing?